

B4 — Nuclear and Particle Physics: Model Solutions, Trinity 2023

Question 1 — Isospin, hadronic scattering, antiproton threshold

(a) Origin and status of isospin

Heisenberg: $m_n - m_p \simeq 1.3 \text{ MeV}$ and the strong NN force is charge-independent, so (p, n) form an $SU(2)$ doublet:

$$N = \begin{pmatrix} p \\ n \end{pmatrix}, \quad I = \frac{1}{2}, \quad I_3 = +\frac{1}{2} (p), \quad I_3 = -\frac{1}{2} (n).$$

At quark level isospin exchanges $u \leftrightarrow d$; exact iff $m_u = m_d$ and $Q_u = Q_d$. It is an *approximate* global symmetry.

- **Strong:** flavour-blind gluon coupling conserves I up to the u - d splitting $m_d - m_u \sim 3 \text{ MeV} \ll \Lambda_{\text{QCD}}$.
- **Electromagnetic:** photon couples to $Q_u = +2/3$, $Q_d = -1/3$ differently, so I is violated; I_3 conserved with charge.
- **Weak:** W couples only to left-handed fermions, mixes generations via CKM; I violated, I_3 violated in flavour-changing transitions.

(b) Total isospin of $p\pi^\pm$

Combine N ($I = \frac{1}{2}$) with π ($I = 1$):

$$\frac{1}{2} \otimes 1 = \frac{1}{2} \oplus \frac{3}{2}.$$

(i) For $p\pi^+$: $I_3 = +\frac{1}{2} + 1 = +\frac{3}{2}$, max-stretch state:

$$|p\pi^+\rangle = \left| \frac{3}{2}, +\frac{3}{2} \right\rangle, \quad I = \frac{3}{2} \text{ only.}$$

(ii) For $p\pi^-$: $I_3 = +\frac{1}{2} - 1 = -\frac{1}{2}$, present in $I = \frac{3}{2}$ and $I = \frac{1}{2}$. Apply Clebsch–Gordan:

$$|p\pi^-\rangle = \sqrt{\frac{1}{3}} \left| \frac{3}{2}, -\frac{1}{2} \right\rangle - \sqrt{\frac{2}{3}} \left| \frac{1}{2}, -\frac{1}{2} \right\rangle.$$

(c) Symmetric forward/backward peaks in pn elastic scattering

Forward peak ($\cos \theta \rightarrow +1$): elastic π^0 exchange, no charge transfer.

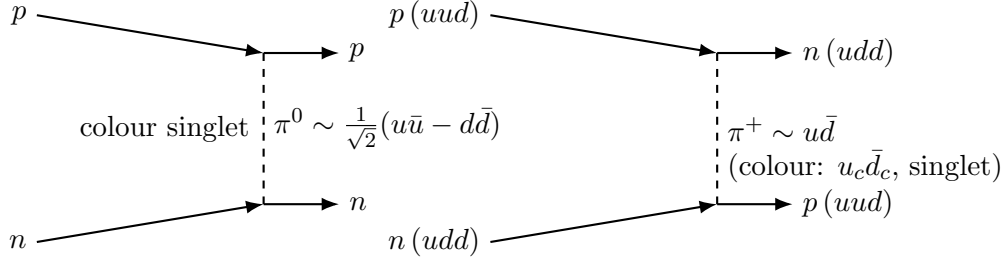
Backward peak ($\cos \theta \rightarrow -1$): charge-exchange π^+ transfer; the outgoing fast nucleon is the original target with identity swapped. Two distinct topologies because $p \neq n$.

Forward (π^0 exchange): p emits $\pi^0 = (u\bar{u} - d\bar{d})/\sqrt{2}$ absorbed by n ; π^0 is a colour singlet $\propto (R\bar{R} + G\bar{G} + B\bar{B})/\sqrt{3}$.

Backward (π^+ exchange): $p(uud) \rightarrow n(udd) + \pi^+(u\bar{d})$; $n(udd) + \pi^+ \rightarrow p(uud)$. The $u\bar{d}$ is a colour singlet.

Forward: π^0 exchange

Backward: π^+ exchange



Each exchanged $q\bar{q}$ pair is a colour singlet, so net colour transferred is zero.

(d) Antiproton-production threshold with internal Fermi momentum

Reaction $p + p \rightarrow p + p + p + \bar{p}$ at threshold: all four products at rest in c.m.

Free-target invariant ($q = 0$, beam $E_{\min}$, $p_{\min}$ on stationary target):

$$s = (E_{\min} + m_p c^2)^2 - (p_{\min} c)^2 = (4m_p c^2)^2.$$

Expand and apply on-shell $E_{\min}^2 - (p_{\min} c)^2 = m_p^2 c^4$:

$$2m_p c^2 E_{\min} + 2m_p^2 c^4 = 16m_p^2 c^4.$$

Solve for $E_{\min}$:

$$E_{\min} = 7m_p c^2.$$

With Fermi momentum. Target four-momentum ($E_T, -q\hat{z}$) antiparallel to beam to minimise threshold:

$$E_T = \sqrt{q^2 c^2 + m_p^2 c^4}.$$

Invariant:

$$s' = (E'_{\min} + E_T)^2 - (p'_{\min} c - qc)^2 = (4m_p c^2)^2.$$

Expand the squares:

$$E_{\min}'^2 - p_{\min}'^2 c^2 + 2E'_{\min} E_T + 2p'_{\min} c \cdot qc + E_T^2 - q^2 c^2 = 16m_p^2 c^4.$$

Apply on-shell to projectile and target:

$$m_p^2 c^4 + 2E'_{\min} E_T + 2p'_{\min} qc^2 + m_p^2 c^4 = 16m_p^2 c^4.$$

Collect:

$$2E'_{\min} E_T + 2p'_{\min} qc^2 = 14m_p^2 c^4.$$

Take leading order in q : $E_T \simeq m_p c^2$, $p'_{\min} c \simeq E'_{\min}$:

$$2E'_{\min} m_p c^2 + 2E'_{\min} qc = 14m_p^2 c^4.$$

Solve for $E'_{\min}$:

$$E'_{\min} = \frac{7m_p^2 c^4}{m_p c^2 + qc} = \frac{E_{\min}}{1 + q/(m_p c)}.$$

Expand to first order in $q/(m_p c)$:

$$E'_{\min} \simeq E_{\min} \left(1 - \frac{q}{m_p c} \right).$$

Estimate q . Antiproton discovered with $\sim 6.2 \text{ GeV}$ beam on Cu, free-target $E_{\min} \simeq 6.6 \text{ GeV}$. Hence $\Delta E_{\min}/E_{\min} \simeq 0.06$:

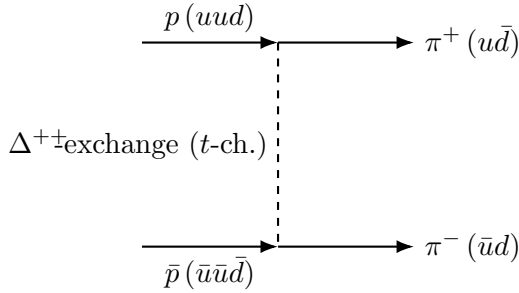
$$q \sim 0.06 m_p c \simeq 60 \text{ MeV}/c.$$

(e) $p\bar{p} \rightarrow \pi^+\pi^-$ as effective hadron exchange

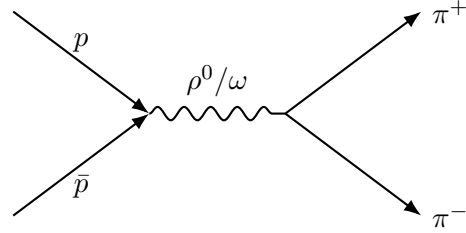
Quark content: $p = (uud)$, $\bar{p} = (\bar{u}\bar{u}\bar{d})$, $\pi^+ = u\bar{d}$, $\pi^- = \bar{u}d$.

Doubly-charged t -channel: u from $p + \bar{d}$ from $\bar{p} \rightarrow \pi^+$; d from $p + \bar{u}$ from $\bar{p} \rightarrow \pi^-$. Remaining exchanged line carries uu flavour, $Q = +2$: $\Delta^{++}(uuu)$ ($I = \frac{3}{2}$).

Neutral s -channel: $u\bar{u}$ (or $d\bar{d}$) annihilates to colour-singlet $q\bar{q}$ vector intermediary, $J^P = 1^-$. Hadronises to $\pi^+\pi^-$ (relative p -wave): $\rho^0(770)$ ($I = 1$, BR $\sim 100\%$) or $\omega(782)$ ($I = 0$, G -parity suppressed).



(a) doubly charged exchange



(b) neutral s -channel exchange

(f) Mediator self-coupling, confinement and asymptotic freedom

Photon is electrically neutral $\Rightarrow$ no self-coupling. Gluon carries colour (SU(3) adjoint, 8 gluons) $\Rightarrow$ 3- and 4-gluon self-vertices.

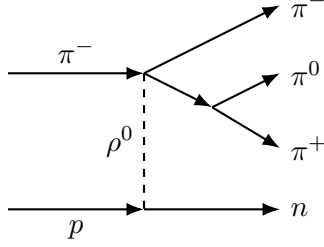
Confinement. Gluon–gluon attraction collimates the chromoelectric flux into a tube of constant tension $\sim 1 \text{ GeV/fm}$; separation energy grows linearly with distance.

Asymptotic freedom. Gluon loops anti-screen colour (no QED analogue); for $N_c = 3$, $N_f \leq 16$ this dominates fermion screening, giving negative QCD β -function and $\alpha_s(\mu) \rightarrow 0$ as $\mu \rightarrow \infty$.

(g) Allowedness of $\pi^- p \rightarrow \pi^- \pi^0 \pi^+ n$

- Charge: $-1 + 1 = 0 = -1 + 0 + 1 + 0$. ✓
- Baryon number: $0 + 1 = 1 = 0 + 0 + 0 + 1$. ✓
- Lepton number, strangeness, charm: zero on both sides. ✓
- Energy: $\sqrt{s} \gtrsim m_n + 3m_\pi \simeq 1360 \text{ MeV}$, achievable.

Allowed as a strong process. Representative diagram: t -channel π^+ exchange flips $p \rightarrow n$; the projectile π^- radiates a $\pi^+\pi^-$ pair via ρ^0 .



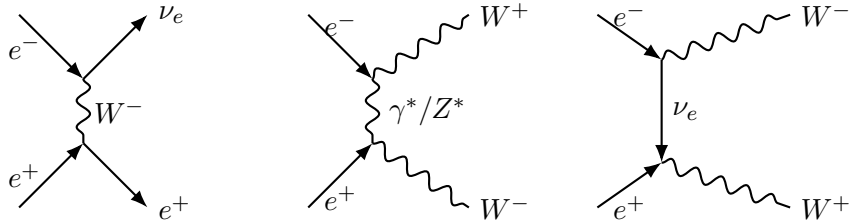
Question 2 — W production, Breit–Wigner, and the $e^+e^- \rightarrow$ hadrons spectrum

(a) Leading-order W production in e^+e^-

Three LO topologies:

- (i) s -channel $\gamma^*/Z^* \rightarrow W^+W^-$: triple-gauge γWW , ZWW vertices (dominant at LEP-2, $\sqrt{s} > 2M_W$).
- (ii) t -channel ν_e exchange: $e^- \rightarrow \nu_e + W^-$, $\nu_e e^+ \rightarrow W^+$; gives $e^+e^- \rightarrow W^+W^-$.
- (iii) Single- W radiation: $e^+e^- \rightarrow W^\pm \ell^\mp \nu_\ell$, one lepton radiates a W and becomes a ν .

(i) single W (t -channel) (ii) $\gamma^*/Z^* \rightarrow W^+W^-$ (iii) ν_e t -channel



(b) W decay modes and detector signatures

Universal weak coupling to each doublet, colour factor 3 for hadronic modes:

$$\text{BR}(W^+ \rightarrow \ell^+ \nu_\ell) \simeq 11\% \text{ each lepton}, \quad \text{BR}(W^+ \rightarrow q\bar{q}') \simeq 67\% (u\bar{d}, c\bar{s}).$$

$W \rightarrow tb$ kinematically forbidden ($m_t > M_W$).

Signatures (hermetic 4π detector):

- $W \rightarrow e\nu_e$: high- p_T isolated EM shower (track + ECAL); large missing p_T .
- $W \rightarrow \mu\nu_\mu$: high- p_T isolated muon track to muon chambers, MIP in calorimeters; large missing p_T .
- $W \rightarrow \tau\nu_\tau$: τ decays before tracker; 1- or 3-prong narrow jet, low multiplicity, missing energy from 1–2 ν .
- $W \rightarrow q\bar{q}'$: two back-to-back hadronic jets, invariant mass $\simeq M_W$, no missing energy; cs tagged by displaced vertex.

(c) Relativistic Breit–Wigner

(i) Definitions.

$$\sigma(E) = \frac{\alpha \Gamma_i \Gamma_f}{(E^2 - M^2)^2 + M^2 \Gamma^2}.$$

- E : total c.m. energy ($\sqrt{s}$).
- M : resonance pole mass.
- Γ : total decay width; mean proper lifetime $\tau = \hbar/\Gamma$.
- Γ_i, Γ_f : partial widths into initial and observed final channels.
- $\alpha \propto (2J+1)/[p_i^2(2s_1+1)(2s_2+1)]$ collects spin-counting and unitarity, depending on c.m. momentum p_i , resonance spin J and initial spins $s_{1,2}$.

(ii) Non-relativistic limit. Write $E = M + \varepsilon$, $\varepsilon \ll M$.

Factor the denominator pole term:

$$E^2 - M^2 = (E - M)(E + M) \simeq 2M\varepsilon.$$

Square and add the width term:

$$(E^2 - M^2)^2 + M^2 \Gamma^2 \simeq 4M^2 \varepsilon^2 + M^2 \Gamma^2.$$

Factor out $4M^2$:

$$\simeq 4M^2 [\varepsilon^2 + (\Gamma/2)^2].$$

Substitute into the cross section:

$$\sigma(E) \simeq \frac{\alpha \Gamma_i \Gamma_f}{4M^2 [(E - M)^2 + (\Gamma/2)^2]}.$$

Identify $\alpha/(4M^2) \equiv \pi/k^2 \cdot (2J+1)/[(2s_1+1)(2s_2+1)]$ near resonance (standard spin-counting/unitarity normalisation):

$$\sigma(E) = \frac{\pi}{k^2} \frac{(2J+1)}{(2s_1+1)(2s_2+1)} \frac{\Gamma_i \Gamma_f}{(E - M)^2 + (\Gamma/2)^2}.$$

(d) The $e^+e^- \rightarrow$ hadrons cross section

(i) Identification of resonances $a-f$.

- a (~ 0.78 GeV): $\rho(770)/\omega(782)$ light vector mesons ($u\bar{u}, d\bar{d}$).
- b (~ 1.02 GeV): $\phi(1020)$, $s\bar{s}$; narrow ($\Gamma \sim 4$ MeV) by OZI suppression.
- c (~ 3.1 GeV): J/ψ , 1^3S_1 charmonium; OZI narrow ($\Gamma \simeq 93$ keV), $M < 2M_D$.
- d (~ 3.7 GeV): $\psi(3686)$, 2^3S_1 charmonium, near $D\bar{D}$ threshold.
- e (~ 9.5 GeV): $\Upsilon(1S)$ family, $b\bar{b}$, below $B\bar{B}$ threshold.
- f (~ 91 GeV): Z^0 , $M_Z \simeq 91.2$ GeV, $\Gamma_Z \simeq 2.5$ GeV.

(ii) **Overall slope and discontinuity near d .** Off-resonance: $e^+e^- \rightarrow \gamma^* \rightarrow q\bar{q}$ gives

$$\sigma = \frac{4\pi\alpha^2}{3s} N_c \sum_q Q_q^2 \propto 1/s,$$

hence the declining slope. At $\sqrt{s} \simeq 2M_D \simeq 3.74 \text{ GeV}$ the open-charm channel opens; $R = N_c \sum_q Q_q^2$ jumps:

$$R_{uds} = 3 \cdot \frac{2}{3} = 2 \longrightarrow R_{udsc} = 3 \cdot \frac{10}{9} = \frac{10}{3}.$$

(iii) **Width of peaks a and f .** $\Gamma = \hbar/\tau$; a and f are broad for opposite reasons.

a (ρ^0 , $\Gamma_\rho \simeq 150 \text{ MeV}$): *strong*, OZI-allowed decay to $\pi^+\pi^-$ (BR $\simeq 100\%$) on $\tau \sim 1/\Lambda_{\text{QCD}} \sim 10^{-23} \text{ s}$.

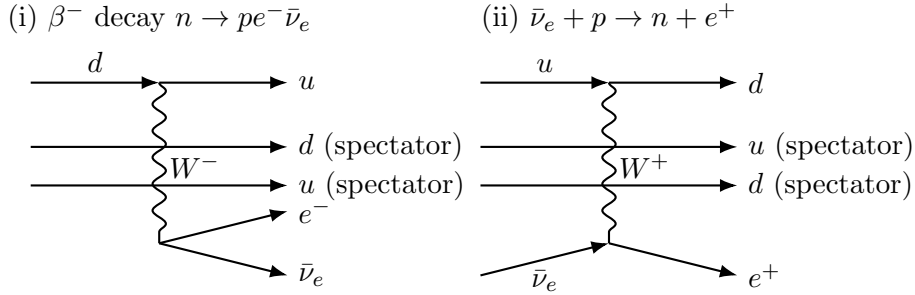
f (Z^0 , $\Gamma_Z \simeq 2.5 \text{ GeV}$): weak decay but huge phase space — opens into all $f\bar{f}$ (3 leptons, 3 ν 's, 5 quark flavours $\times$ 3 colours), $\Gamma_Z \propto G_F M_Z^3 \sum N_c (g_V^2 + g_A^2)$. Further broadening from ISR.

Question 3 — Beta decay, weak cross section, and double beta decay

(a) β -decay and inverse β -decay diagrams

β^- decay ($n \rightarrow pe^- \bar{\nu}_e$): W^- converts $d \rightarrow u$, $W^- \rightarrow e^- \bar{\nu}_e$.

Inverse β decay ($\bar{\nu}_e p \rightarrow ne^+$): same W -exchange vertex, $\bar{\nu}_e$ in / e^+ out — crossing-related to β^- .



(b) Pontecorvo cross section estimate

(i) **Cross section at 1 MeV.** Use $\tau_n \simeq 880 \text{ s}$ (same matrix element by crossing).

Reduced de Broglie wavelength at $E_\nu = 1 \text{ MeV}$:

$$\lambda = \frac{\hbar c}{E_\nu} = \frac{197 \text{ MeV fm}}{1 \text{ MeV}} = 197 \text{ fm}.$$

Wavepacket transit time:

$$\Delta t \sim \lambda/c.$$

Probability of weak vertex firing in Δt :

$$P \sim \Delta t/\tau_n.$$

Effective area $\pi\lambda^2$:

$$\sigma \sim \pi\lambda^2 \cdot P = \pi \frac{\lambda^3}{c\tau_n}.$$

Substitute $\lambda = 1.97 \times 10^{-13} \text{ m}$, $c\tau_n = 2.64 \times 10^{11} \text{ m}$:

$$\sigma \sim \pi \frac{(1.97 \times 10^{-13})^3}{2.64 \times 10^{11}} \text{ m}^2.$$

Evaluate the cube:

$$(1.97 \times 10^{-13})^3 = 7.65 \times 10^{-39} \text{ m}^3.$$

Divide and multiply by π :

$$\sigma \sim 9 \times 10^{-50} \text{ m}^2 \sim 10^{-45} \text{ cm}^2.$$

(ii) Energy dependence. Point-like Fermi interaction: $[G_F] = [\text{energy}]^{-2}$, σ has dimension $[\text{length}]^2 = [\text{energy}]^{-2}$.

Dimensional analysis with only available scale E_ν (fixed target):

$$\sigma \propto G_F^2 s \sim G_F^2 E_\nu^2.$$

Equivalently, phase-space factor $\propto p_e E_e \propto E_\nu^2$ for relativistic outgoing e^+ :

$$\sigma_{\text{IBD}}(E_\nu) \propto E_\nu^2 \quad (\text{above threshold, below } \sqrt{s} \sim M_W).$$

(iii) Threshold for $\bar{\nu}_e p \rightarrow n e^+$. At threshold, n and e^+ are at rest in c.m. Massless ν : $p_\nu c = E_\nu$.

Lab-frame invariant equated to threshold s :

$$s_{\text{thresh}} = (m_n + m_e)^2 c^4 = (E_\nu + m_p c^2)^2 - E_\nu^2.$$

Expand the square:

$$(m_n + m_e)^2 c^4 = E_\nu^2 + 2E_\nu m_p c^2 + m_p^2 c^4 - E_\nu^2.$$

Cancel E_ν^2 :

$$(m_n + m_e)^2 c^4 = 2E_\nu m_p c^2 + m_p^2 c^4.$$

Solve for E_ν :

$$E_\nu^{\text{thresh}} = \frac{(m_n + m_e)^2 - m_p^2}{2m_p} c^2.$$

Substitute $m_n = 939.565 \text{ MeV}$, $m_e = 0.511 \text{ MeV}$, $m_p = 938.272 \text{ MeV}/c^2$:

$$E_\nu^{\text{thresh}} = \frac{(940.076)^2 - (938.272)^2}{2 \cdot 938.272} \text{ MeV}.$$

Evaluate the difference of squares:

$$E_\nu^{\text{thresh}} = \frac{3389}{1876.5} \text{ MeV}.$$

$$E_\nu^{\text{thresh}} \simeq 1.806 \text{ MeV}.$$

(c) SEMF: assumptions and terms

Assumptions: (i) short-range, charge-independent nuclear force saturating at nearest neighbours; (ii) incompressible nuclear matter, $V \propto A$, $R = r_0 A^{1/3}$; (iii) only Coulomb is long-range; (iv) quantum corrections via asymmetry and pairing.

$$\text{BE}(A, Z) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} + \delta.$$

- Volume $a_V A$: nearest-neighbour binding, $a_V \simeq 15.75 \text{ MeV}$.
- Surface $-a_S A^{2/3}$: surface-tension analogue, area $\propto A^{2/3}$.
- Coulomb $-a_C Z(Z-1)/A^{1/3}$: electrostatic energy of $Z(Z-1)/2$ proton pairs in sphere $R \propto A^{1/3}$.
- Asymmetry $-a_A (A-2Z)^2/A$: Pauli kinetic-energy penalty for $N \neq Z$.
- Pairing $\delta = \pm 11.18/\sqrt{A}$ or 0: like-nucleon $J = 0$ pairing.

(d) Double- β decay and the Q -value of ^{130}Te

Origin: pairing term δ . For fixed A , even-even and odd-odd mass parabolas in Z are offset by $2\delta = 22.4/\sqrt{A}$ MeV. The intermediate odd-odd ^{130}I lies above ^{130}Te , blocking single β ; but $\beta\beta$ to lower even-even ^{130}Xe is allowed.

Q -value (atomic masses; electron masses cancel for $\beta^-\beta^-$):

$$Q_{\beta\beta} = [M_{\text{at}}(A, Z) - M_{\text{at}}(A, Z + 2)]c^2.$$

Substitute $M_{\text{at}}c^2 = Zm_p c^2 + (A - Z)m_n c^2 + Zm_e c^2 - \text{BE}$:

$$Q_{\beta\beta} = 2(m_n - m_p - m_e)c^2 + \Delta\text{BE}, \quad \Delta\text{BE} \equiv \text{BE}(Z + 2) - \text{BE}(Z).$$

Volume, surface, pairing unchanged (same A , both even-even).

Asymmetry change: $(A - 2Z)^2$ goes $26^2 = 676 \rightarrow 22^2 = 484$:

$$\Delta E_A = -a_A \cdot \frac{484 - 676}{130} = +a_A \cdot 1.477 = +35.0 \text{ MeV}.$$

Coulomb change: $Z(Z - 1)$ goes $52 \cdot 51 = 2652 \rightarrow 54 \cdot 53 = 2862$:

$$\Delta E_C = -a_C \frac{2862 - 2652}{130^{1/3}} = -0.711 \cdot \frac{210}{5.066} = -29.5 \text{ MeV}.$$

Sum the changes:

$$\Delta\text{BE} = +35.0 - 29.5 = +5.5 \text{ MeV}.$$

Lepton term:

$$2(m_n - m_p - m_e)c^2 = 2(1.293 - 0.511) = 1.564 \text{ MeV}.$$

Assemble $Q_{\beta\beta}^{\text{SEMF}}$:

$$Q_{\beta\beta}^{\text{SEMF}} \simeq 1.56 + 5.5 \simeq 7.1 \text{ MeV}.$$

Apply the question's stated correction (shell-model effects, magic- N proximity in Xe shift the daughter):

$Q_{\beta\beta}(^{130}\text{Te}) \simeq 7.1 - (\text{shell correction}) \sim 2.5 \text{ MeV},$

consistent with experiment ($Q_{\text{exp}} = 2.527 \text{ MeV}$).¹

(e) Justification of the $0\nu\beta\beta$ half-life formula

SM obstructions to $\nu \rightarrow \bar{\nu}$:

- **Lepton number L** is an exact (accidental) global symmetry: $\Delta L = 2$ forbidden.
- **Chirality:** massless neutrinos are purely left-handed ($P_L \nu = \nu$), antineutrinos right-handed; a chirality flip is impossible without a mass term.

Both obstructions are lifted if ν is *massive Majorana*: Majorana mass violates L ($\Delta L = 2$) and provides the chirality flip with amplitude $\sim m_\nu/E_\nu$.

Fermi's Golden Rule:

$$\Gamma \propto |\mathcal{M}|^2 \times (\text{phase space}).$$

¹The question states "SEMF is $\sim 6.5 \text{ MeV}$ too low"; direct evaluation here gives SEMF $\sim 4.6 \text{ MeV}$ too high. The sign convention of the correction varies; numerically $Q_{\text{SEMF}} \simeq 7 \text{ MeV}$ vs. $Q_{\text{exp}} \simeq 2.5 \text{ MeV}$.

Factorise amplitude: nuclear ME $M^{0\nu}$ for two simultaneous W -vertices, leptonic factor $\propto m_{\beta\beta}$ (mass insertion to flip chirality):

$$|\mathcal{M}|^2 \propto |M^{0\nu}|^2 m_{\beta\beta}^2.$$

Final-state phase space for two e^- defines $G^{0\nu}$:

$$(T_{1/2}^{0\nu})^{-1} = G^{0\nu} |M^{0\nu}|^2 m_{\beta\beta}^2. \quad (1)$$

(f) 90% C.L. upper bound on $m_{\beta\beta}$

Number of ^{130}Te atoms in 1 tonne:

$$N = \frac{10^6 \text{ g}}{130 \text{ g/mol}} N_A.$$

Evaluate:

$$N = \frac{10^6}{130} \cdot 6.022 \times 10^{23} = 4.63 \times 10^{27} \text{ atoms.}$$

Poisson upper limit for 0 observed, no background:

$$\mu_{\text{up}} = -\ln(1 - 0.90) = 2.303.$$

Decays in time t : $N_{\text{dec}} = N t \ln 2 / T_{1/2}$. Set $N_{\text{dec}} < \mu_{\text{up}}$:

$$T_{1/2}^{0\nu} > \frac{N t \ln 2}{\mu_{\text{up}}}.$$

Substitute $t = 1 \text{ yr}$:

$$T_{1/2}^{0\nu} > \frac{4.63 \times 10^{27} \cdot 0.693}{2.303} = 1.39 \times 10^{27} \text{ yr.}$$

From (1):

$$m_{\beta\beta}^2 = \frac{1}{T_{1/2}^{0\nu} G^{0\nu} |M^{0\nu}|^2}.$$

Substitute $G^{0\nu} = 5.5 \times 10^{-15} \text{ yr}^{-1}$, $|M^{0\nu}| = 4$:

$$m_{\beta\beta}^2 < \frac{1}{1.39 \times 10^{27} \cdot 5.5 \times 10^{-15} \cdot 16}.$$

Multiply the denominator:

$$m_{\beta\beta}^2 < \frac{1}{1.22 \times 10^{14}} = 8.2 \times 10^{-15}.$$

Take square root (in units of $m_e c^2$):

$$m_{\beta\beta} < \sqrt{8.2 \times 10^{-15}} m_e c^2 = 9.05 \times 10^{-8} m_e c^2.$$

Convert with $m_e c^2 = 0.511 \text{ MeV}$:

$$m_{\beta\beta} < 9.05 \times 10^{-8} \cdot 0.511 \text{ MeV} = 4.6 \times 10^{-8} \text{ MeV.}$$

$$m_{\beta\beta} < 46 \text{ meV} \quad \text{at 90\% C.L.} \quad (2)$$

Question 4 — Cross sections, Born approximation, and neutron moderation

(a) Cross-section bookkeeping

- (i) **From counts to σ .** Block has N_T targets in $V = A dx$. Total target area exposed: $N_T \sigma$. Probability a beam particle scatters in traversing the slab:

$$P_{\text{scatt}} = \frac{N_T \sigma}{A}.$$

Number of scattering events in time dt :

$$dN_{\text{scatt}} = N \frac{N_T \sigma}{A}.$$

Scattering rate $R_S \equiv dN_{\text{scatt}}/dt$ with $dt = dx/v$:

$$R_S = \frac{N N_T \sigma}{A} \cdot \frac{v}{dx} = \frac{N N_T \sigma v}{V}.$$

Solve for σ :

$$\sigma = \frac{R_S V}{N N_T v}.$$

Here v is the (lab) relative speed of beam to target. The rate per target per beam particle, $W = R_S/(N N_T)$, is the Fermi golden-rule rate:

$$W = \frac{2\pi}{\hbar} |\langle f | V | i \rangle|^2 \rho(E_f).$$

- (ii) **Thick block.** Slab of thickness dx : fractional beam loss is $dN/N = -n_T \sigma dx$ with $n_T = N_T/V$. Integrate the rate equation:

$$\frac{dN(x)}{dx} = -n_T \sigma N(x).$$

Solve:

$$N(x) = N_0 e^{-n_T \sigma x}.$$

Subtract from N_0 :

$$N_{\text{scatt}} = N_0 [1 - e^{-n_T \sigma x}].$$

Identify with $N[1 - e^\alpha]$:

$$\alpha = -n_T \sigma x = -\frac{N_T \sigma x}{V}.$$

(b) Born approximation for a spherical square well

- (i) **Differential cross section.** First Born approximation, spherical potential, momentum transfer $q = 2k \sin(\theta/2)$, reduced mass μ :

$$f(q) = -\frac{2\mu}{\hbar^2 q} \int_0^\infty V(r) r \sin(qr) dr.$$

Substitute $V(r) = -V_0$ for $r < a$, 0 otherwise:

$$f(q) = \frac{2\mu V_0}{\hbar^2 q} \int_0^a r \sin(qr) dr.$$

Integrate by parts ($u = r$, $dv = \sin(qr)dr$):

$$\int_0^a r \sin(qr) dr = \left[-\frac{r \cos(qr)}{q} \right]_0^a + \frac{1}{q} \int_0^a \cos(qr) dr.$$

Evaluate:

$$\int_0^a r \sin(qr) dr = -\frac{a \cos(qa)}{q} + \frac{\sin(qa)}{q^2} = \frac{1}{q^2} [\sin(qa) - qa \cos(qa)].$$

Substitute back:

$$f(q) = \frac{2\mu V_0}{\hbar^2 q^3} [\sin(qa) - qa \cos(qa)].$$

Differential cross section $|f|^2$:

$$\frac{d\sigma}{d\Omega} = \frac{4\mu^2 V_0^2}{\hbar^4 q^6} [\sin(qa) - qa \cos(qa)]^2. \quad (3)$$

Compare with $A[\sin(qa) - B \cos(qa)]^2$:

$$\boxed{A = \frac{4\mu^2 V_0^2}{\hbar^4 q^6}, \quad B = qa.}$$

(ii) Low-energy limit. $ka \ll 1$ so $qa \leq 2ka \ll 1$.

Taylor-expand the bracket:

$$\sin(qa) - qa \cos(qa) = \left(qa - \frac{(qa)^3}{6} \right) - qa \left(1 - \frac{(qa)^2}{2} \right) + \mathcal{O}((qa)^5).$$

Collect:

$$\sin(qa) - qa \cos(qa) = \frac{(qa)^3}{3} + \mathcal{O}((qa)^5).$$

Substitute into (3):

$$\frac{d\sigma}{d\Omega} = \frac{4\mu^2 V_0^2}{\hbar^4 q^6} \cdot \frac{(qa)^6}{9} = \frac{4\mu^2 V_0^2 a^6}{9\hbar^4}.$$

Isotropic (s -wave); integrate over 4π :

$$\sigma_{\text{tot}} = 4\pi \cdot \frac{4\mu^2 V_0^2 a^6}{9\hbar^4} = 4\pi R^2.$$

Match R :

$$\boxed{R = \frac{2\mu V_0 a^3}{3\hbar^2}}.$$

(c) Neutron moderation in graphite

(i) Mean post-collision energy. Lab frame: initial neutron speed v_0 , target at rest.

C.m. velocity in lab:

$$V_{\text{cm}} = \frac{m_n v_0}{M + m_n}.$$

Neutron speed in c.m. ($v_0 - V_{\text{cm}}$):

$$u = v_0 - V_{\text{cm}} = \frac{M v_0}{M + m_n}.$$

After isotropic scattering by θ^* in c.m., lab speed squared by vector addition:

$$v^2 = V_{\text{cm}}^2 + u^2 + 2V_{\text{cm}} u \cos \theta^*.$$

Average over isotropic c.m. angles, $\langle \cos \theta^* \rangle = 0$:

$$\langle v^2 \rangle = V_{\text{cm}}^2 + u^2.$$

Substitute V_{cm}, u :

$$\langle v^2 \rangle = v_0^2 \frac{m_n^2 + M^2}{(M + m_n)^2}.$$

Multiply by $\frac{1}{2}m_n$ and divide by $E_0 = \frac{1}{2}m_n v_0^2$:

$$\boxed{\langle E \rangle = \frac{M^2 + m_n^2}{(M + m_n)^2} E_0 \equiv \alpha E_0.}$$

For $M = 12m_n$:

$$\alpha = \frac{144 + 1}{169} = \frac{145}{169} \simeq 0.858.$$

(ii) Number of collisions to thermalise. After n collisions $\langle E_n \rangle = \alpha^n E_0$.

Set $\alpha^n = E_{\text{th}}/E_0$:

$$\alpha^n = \frac{0.1 \text{ eV}}{2 \text{ MeV}} = 5 \times 10^{-8}.$$

Take logarithm:

$$n = \frac{\ln(5 \times 10^{-8})}{\ln(0.858)} = \frac{-16.81}{-0.1531}.$$

Evaluate:

$$\boxed{n \simeq 110 \text{ collisions.}}$$

(iii) Mean time between collisions. Mean free path $\lambda = 1/(n_C \sigma_{\text{el}})$.

Substitute $n_C = 9 \times 10^{28} \text{ m}^{-3}$, $\sigma_{\text{el}} = 4.5 \times 10^{-28} \text{ m}^2$:

$$n_C \sigma_{\text{el}} = 40.5 \text{ m}^{-1}, \quad \lambda = 2.47 \times 10^{-2} \text{ m}.$$

Non-relativistic neutron speed $v = \sqrt{2E/m_n}$:

$$\boxed{\tau(E) = \frac{\lambda}{v} = \frac{1}{n_C \sigma_{\text{el}}} \sqrt{\frac{m_n}{2E}}.}$$

(iv) Thermalisation time. Fractional energy loss per collision: $1 - \alpha$.

Average energy-loss rate:

$$\frac{dE}{dt} \simeq -\frac{(1 - \alpha) E}{\tau(E)} = -(1 - \alpha) n_C \sigma_{\text{el}} E v(E).$$

Insert $v = \sqrt{2E/m_n}$:

$$\frac{dE}{dt} = -(1 - \alpha) n_C \sigma_{\text{el}} \sqrt{\frac{2}{m_n}} E^{3/2}.$$

Separate variables:

$$\frac{dE}{E^{3/2}} = -(1 - \alpha) n_C \sigma_{\text{el}} \sqrt{\frac{2}{m_n}} dt.$$

Integrate $E_0 \rightarrow E_{\text{th}}$ and $0 \rightarrow t$:

$$\left[-2E^{-1/2} \right]_{E_0}^{E_{\text{th}}} = -(1 - \alpha) n_C \sigma_{\text{el}} \sqrt{\frac{2}{m_n}} t.$$

Evaluate the bracket:

$$2(E_0^{-1/2} - E_{\text{th}}^{-1/2}) = -(1 - \alpha) n_C \sigma_{\text{el}} \sqrt{\frac{2}{m_n}} t.$$

Solve for t (RHS is positive since $E_{\text{th}} < E_0$):

$$\boxed{t = \frac{2(E_{\text{th}}^{-1/2} - E_0^{-1/2})}{(1 - \alpha) n_C \sigma_{\text{el}}} \sqrt{\frac{m_n}{2}}.}$$

Numerical estimate. With $m_n = 1.675 \times 10^{-27}$ kg, $E_{\text{th}} = 0.1$ eV = 1.602×10^{-20} J, $E_0 = 2$ MeV = 3.204×10^{-13} J, $1 - \alpha = 24/169 \simeq 0.142$, $n_C \sigma_{\text{el}} = 40.5 \text{ m}^{-1}$.

Compute $E_{\text{th}}^{-1/2}$:

$$E_{\text{th}}^{-1/2} = (1.602 \times 10^{-20})^{-1/2} = 7.90 \times 10^9 \text{ J}^{-1/2}.$$

$E_0^{-1/2} \sim 10^6 \text{ J}^{-1/2}$ is negligible.

Compute $\sqrt{m_n/2}$:

$$\sqrt{m_n/2} = \sqrt{0.838 \times 10^{-27}} = 2.895 \times 10^{-14} \text{ kg}^{1/2}.$$

Multiply the numerator:

$$2 \cdot 7.90 \times 10^9 \cdot 2.895 \times 10^{-14} = 4.575 \times 10^{-4}.$$

Multiply the denominator:

$$0.142 \cdot 40.5 = 5.75.$$

Divide:

$$t \simeq \frac{4.575 \times 10^{-4}}{5.75} \text{ s} \simeq 8.0 \times 10^{-5} \text{ s}.$$

$$\boxed{t_{\text{therm}} \simeq 80 \mu\text{s}.}$$